UNIVERSITATEA TEHNICĂ A MOLDOVEI

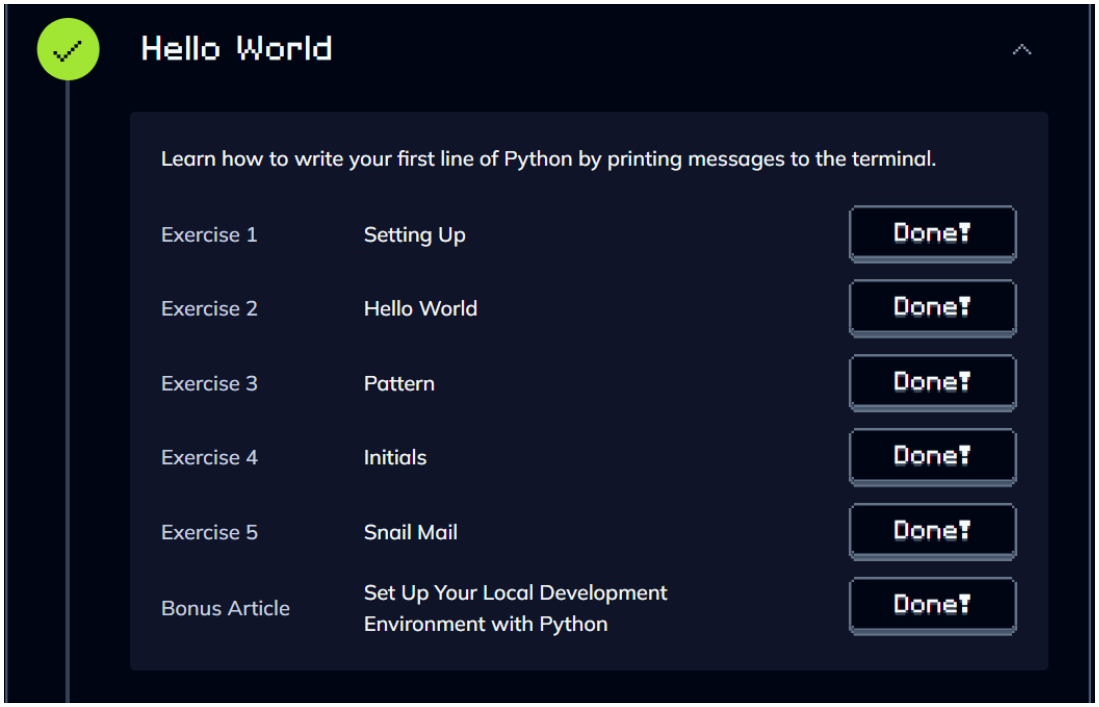
FACULTATEA CALCULATOARE, INFORMATICĂ ȘI MICROELECTRONICĂ

DEPARTAMENTUL INGINERIA SOFTWARE ȘI AUTOMATICĂ

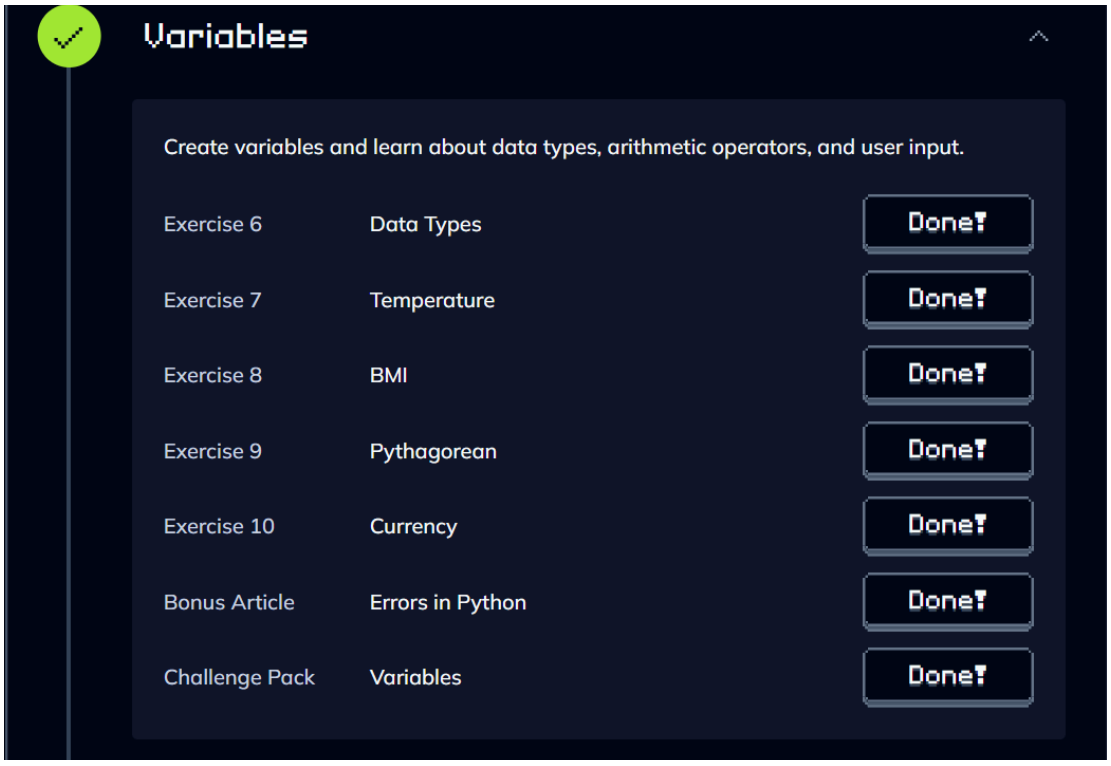
Lucrare de laborator nr. 4
Tehnici de programare

Realizat de : Bulat Madalina
st. gr. SI-262
Verificat de : Emil Cebotaros
asist. univ.

Chisinau 2026



Imaginea 1.1



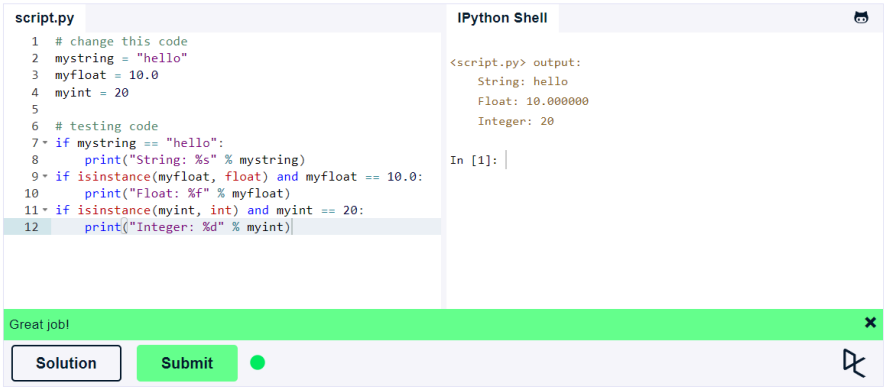
Imaginea 1.2

Use the "print" function to print the line "Hello, World!".



Imaginea 2.1

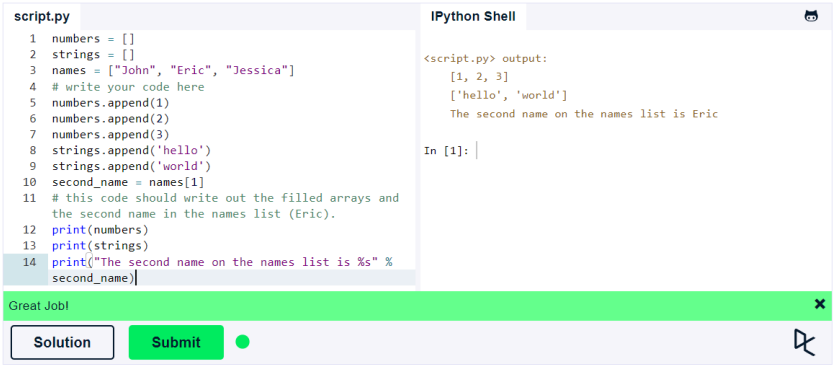
The target of this exercise is to create a string, an integer, and a floating point number. The string should be named `mystring` and should contain the word "hello". The floating point number should be named `myfloat` and should contain the number 10.0, and the integer should be named `myint` and should contain the number 20.



Imaginea 2.2

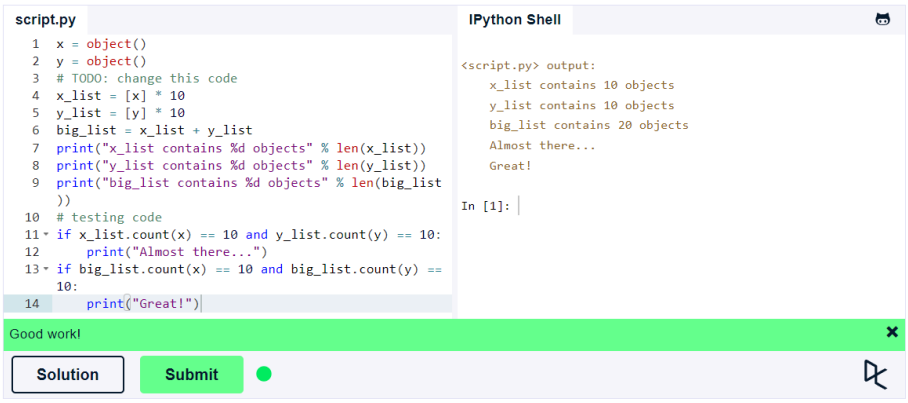
In this exercise, you will need to add numbers and strings to the correct lists using the "append" list method. You must add the numbers 1,2, and 3 to the "numbers" list, and the words 'hello' and 'world' to the strings variable.

You will also have to fill in the variable second_name with the second name in the names list, using the brackets operator `[]`. Note that the index is zero-based, so if you want to access the second item in the list, its index will be 1.



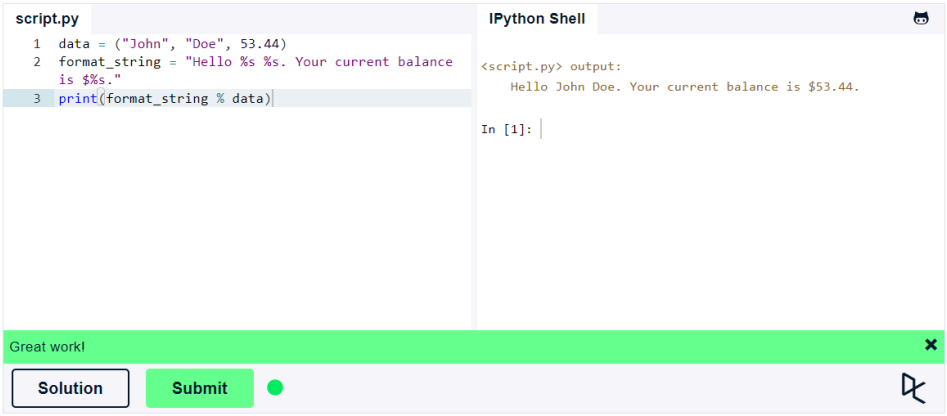
Imaginea 2.3

The target of this exercise is to create two lists called `x_list` and `y_list`, which contain 10 instances of the variables `x` and `y`, respectively. You are also required to create a list called `big_list`, which contains the variables `x` and `y`, 10 times each, by concatenating the two lists you have created.



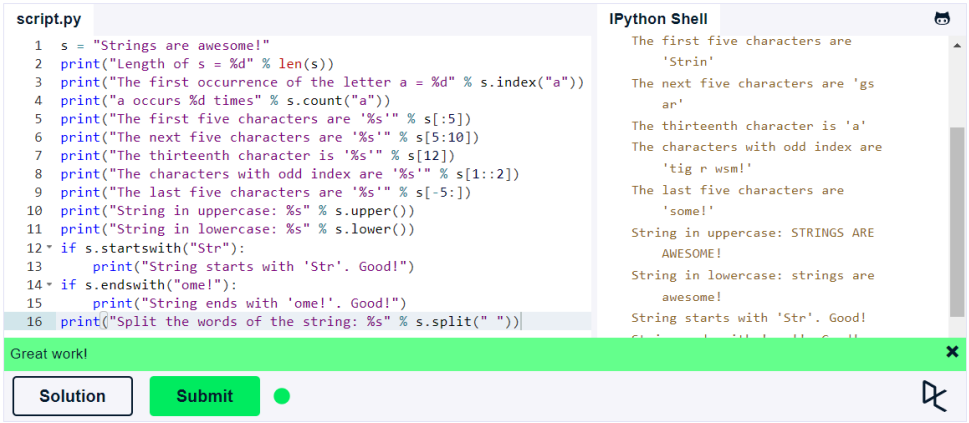
Imaginea 2.4

You will need to write a format string which prints out the data using the following syntax: `Hello John Doe. Your current balance is $53.44.`



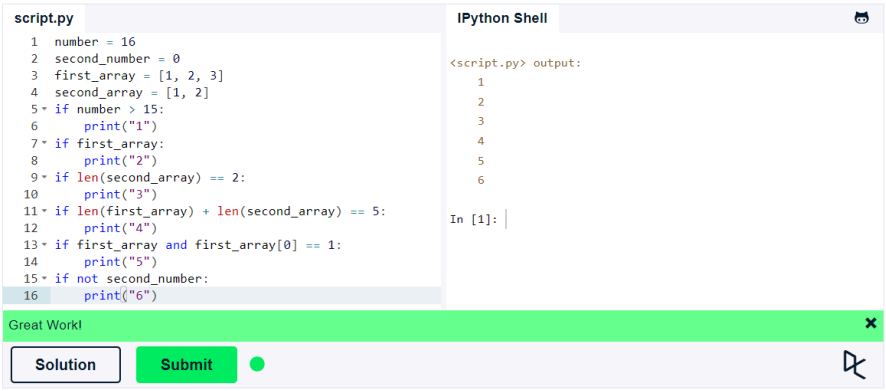
Imaginea 2.5

Try to fix the code to print out the correct information by changing the string.



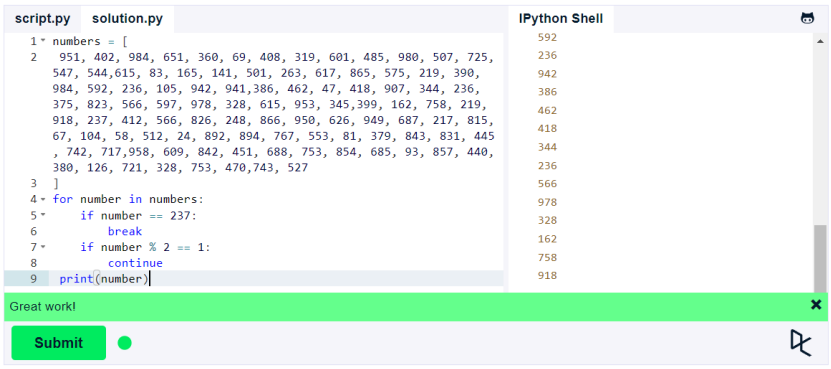
Imaginea 2.6

Change the variables in the first section, so that each if statement resolves as True.



Imaginea 2.7

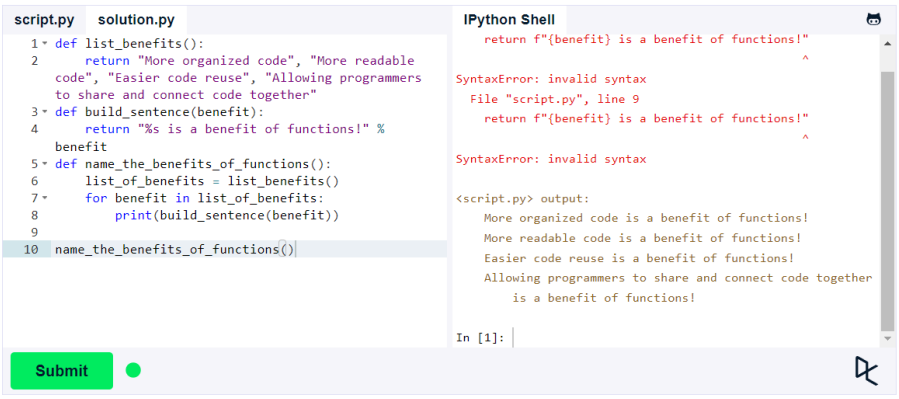
Loop through and print out all even numbers from the numbers list in the same order they are received. Don't print any numbers that come after 237 in the sequence.



Imaginea 2.8

In this exercise you'll use an existing function, and while adding your own to create a fully functional program.

1. Add a function named `list_benefits()` that returns the following list of strings: "More organized code", "More readable code", "Easier code reuse", "Allowing programmers to share and connect code together"
2. Add a function named `build_sentence(info)` which receives a single argument containing a string and returns a sentence starting with the given string and ending with the string " is a benefit of functions!"
3. Run and see all the functions work together!



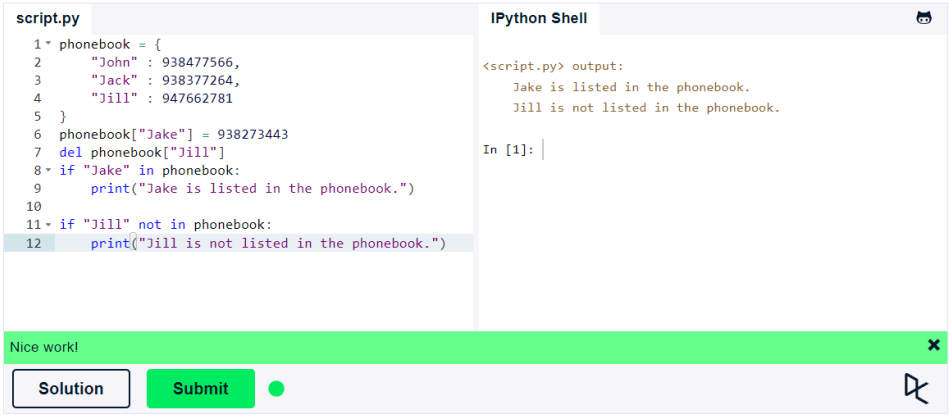
Imaginea 2.9

We have a class defined for vehicles. Create two new vehicles called car1 and car2. Set car1 to be a red convertible worth $60,000.00 with a name of Fer, and car2 to be a blue van named Jump worth $10,000.00.



Imaginea 2.10

Add "Jake" to the phonebook with the phone number 938273443, and remove Jill from the phonebook.



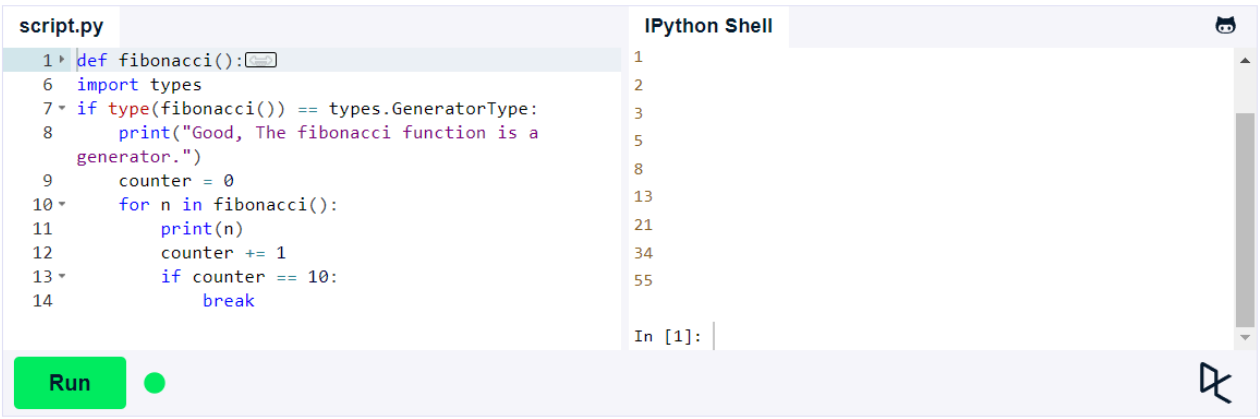
Imaginea 2.11

In this exercise, print an alphabetically sorted list of all the functions in the re module containing the word find.



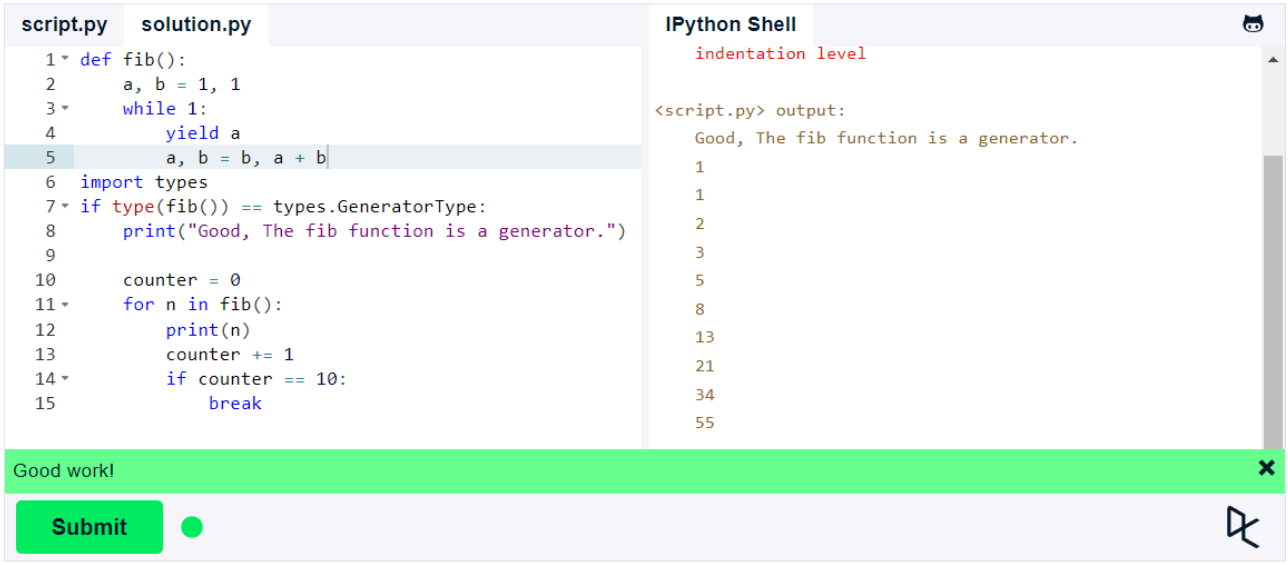
Imaginea 2.12

Write a generator function which returns the Fibonacci series. They are calculated using the following formula: The first two numbers of the series is always equal to 1, and each consecutive number returned is the sum of the last two numbers. Hint: Can you use only two variables in the generator function? Remember that assignments can be done simultaneously. The code



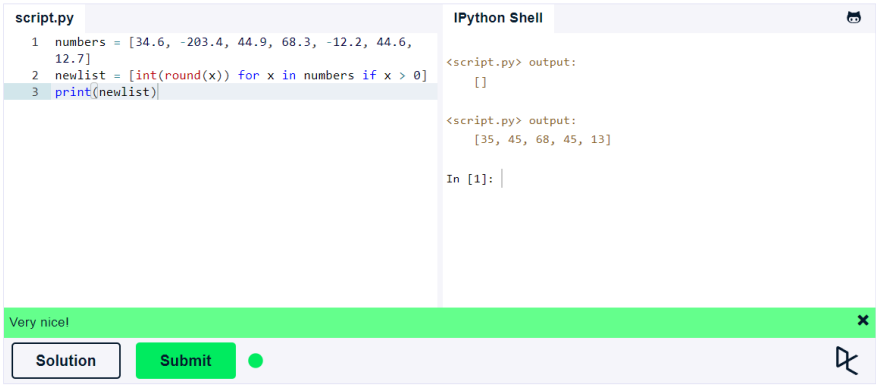
Imaginea 2.13

will simultaneously switch the values of a and b.



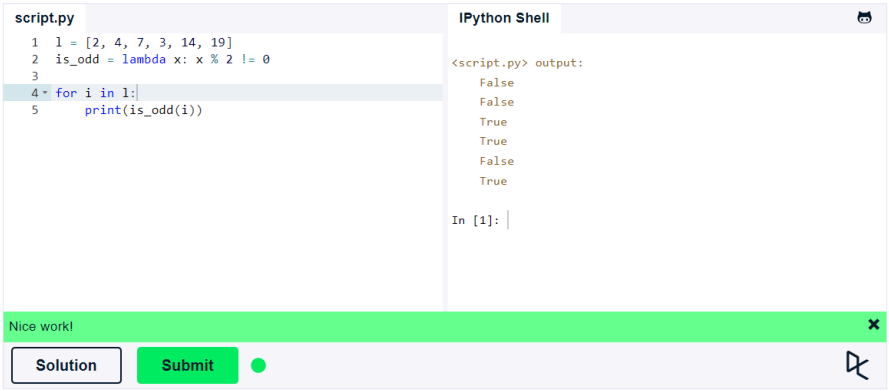
Imaginea 2.14

Using a list comprehension, create a new list called "newlist" out of the list "numbers", which contains only the positive numbers from the list, as integers.



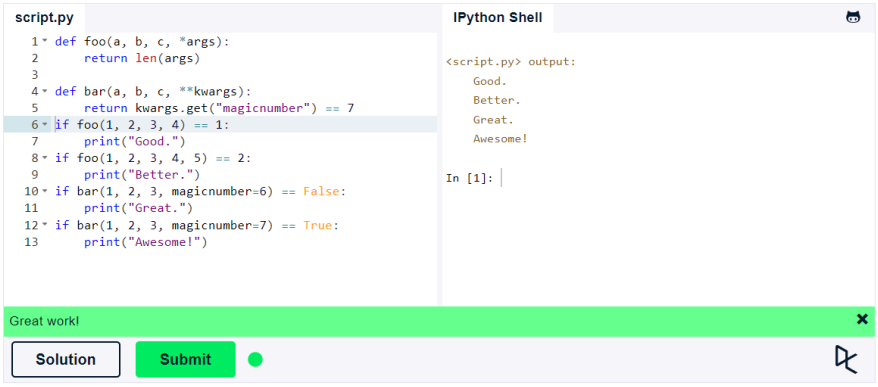
Imaginea 2.15

Write a program using lambda functions to check if a number in the given list is odd. Print "True" if the number is odd or "False" if not for each element.



Imaginea 2.16

Fill in the foo and bar functions so they can receive a variable amount of arguments (3 or more) The foo function must return the amount of extra arguments received. The bar must return True if the argument with the keyword magicnumber is worth 7, and False otherwise.



Imaginea 2.17

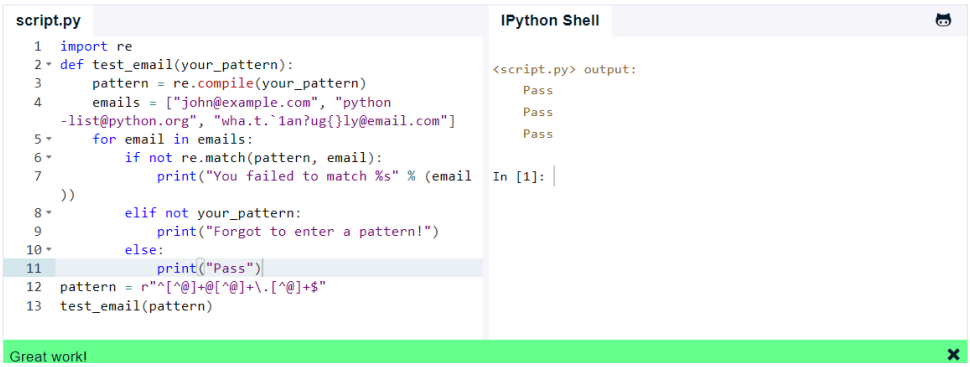
Regular Expressions (sometimes shortened to regexp, regex, or re) are a tool for matching patterns in text. In Python, we have the re module. The applications for regular expressions are wide-spread, but they are fairly complex, so when contemplating using a regex for a certain task, think about alternatives, and come to regexes as a last resort.

An example regex is `r"^(From|To|Cc).*?python-list@python.org"` Now for an explanation: the caret `^` matches text at the beginning of a line. The following group, the part with `(From|To|Cc)` means that the line has to start with one of the words that are separated by the pipe `|`. That is called the OR operator, and the regex will match if the line starts with any of the words in the group. The `.*?` means to un-greedily match any number of characters, except the newline `\n` character. The un-greedy part means to match as few repetitions as possible. The `.` character means any non-newline character, the `*` means to repeat 0 or more times, and the `?` character makes it un-greedy.

So, the following lines would be matched by that regex: `From: python-list@python.org To: !asp]<,. python-list@python.org`

A complete reference for the re syntax is available at the python docs.

As an example of a "proper" email-matching regex (like the one in the exercise), see this



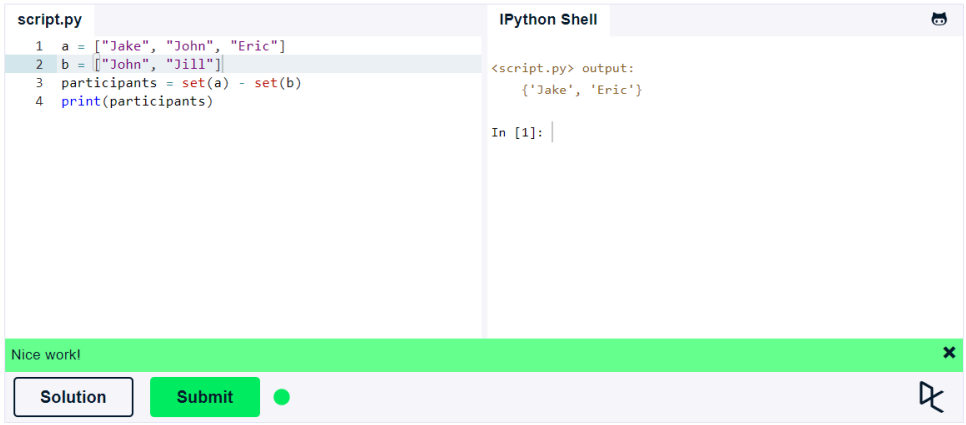
Imaginea 2.18

Handle all the exception! Think back to the previous lessons to return the last name of the actor.



Imaginea 2.19

In the exercise below, use the given lists to print out a set containing all the participants from event A which did not attend event B.



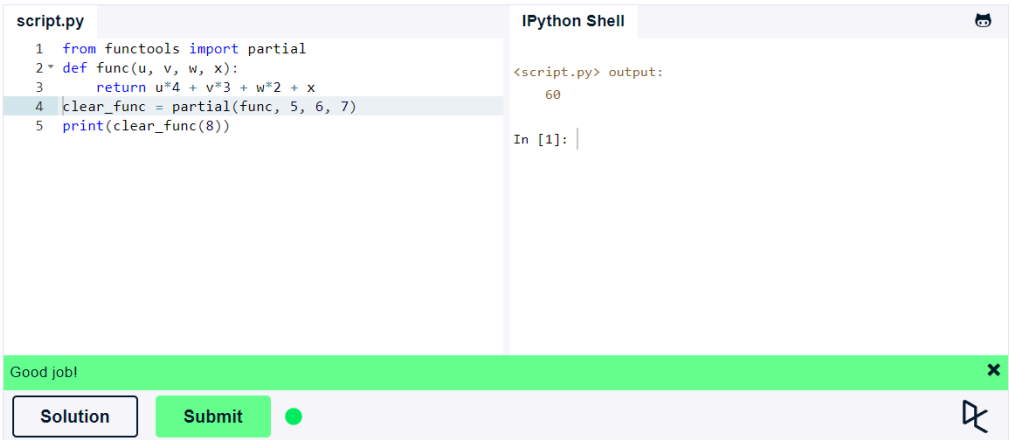
Imaginea 2.20

The aim of this exercise is to print out the JSON string with key-value pair "Me" : 800 added to it.



Imaginea 2.21

Edit the function provided by calling partial() and replacing the first three variables in func(). Then print with the new partial function using only one input variable so that the output equals 60.



Imaginea 2.22

Concluzii

În cadrul lucrării de laborator nr. 4, am învățat noțiunile de bază ale limbajului de programare Python, rezolvând exerciții practice pe platformele CodeDex.io și LearnPython.org. Am exersat scrierea și executarea codului, utilizarea variabilelor, a tipurilor de date, a operatorilor și a structurilor de control. Prin rezolvarea exercițiilor, mi-am dezvoltat gândirea logică și capacitatea de a identifica și corecta erorile. Această lucrare m-a ajutat să-mi consolidez cunoștințele de programare și să înțeleg mai bine bazele limbajului Python.

Bibliografie

https://www.codedex.io

https://www.learnpython.org